

---

# Pedersen’s Haar-averaged fidelity for the TriQG OR and CCX gates

why the  $\pm$ -amplitude pulse design does *not* make it equal to the basis average

---

## TriQG internal note<sup>1</sup>

<sup>1</sup>TriQG/reference/pedersen\_average\_fidelity\_TriQG\_gates.{typ,pdf}

April 2026

### ABSTRACT

This note records exactly what calculation was done when we replaced the computational-basis average  $\overline{F}_{\text{basis}} = \frac{1}{8} \sum_k F_k$  with the Haar-averaged fidelity of [1] for the three-qubit OR and CCX gates simulated in `examples/Average_fidelity_Vcc_newenergy/`. It also explains why the raw Pedersen number ( $\overline{F}_{\text{OR}} = 0.1806$ ,  $\overline{F}_{\text{CCX}} = 0.2820$ ) is so much smaller than the basis average (0.9926 and 0.9966) even though our pulse design *already* uses the  $\pm$ -amplitude trick to cancel the single-atom Rabi dynamical phase. The short answer: the  $\pm$  trick cancels the *single-atom* dynamical phase on a control that completes a full Rabi round-trip, but it does *nothing* to cancel (i) the  $V_{cc}$  static phase acquired while the two controls sit in  $|rr\rangle$  during the target-pulse wait window, (ii) the AC-Stark / Rabi-cycle phase picked up by the blockaded target, or (iii) the  $(-1)$  factor left behind by three sequential target  $\pi$ -pulses of the same sign. These three coherent phase channels produce branch-dependent phases  $\varphi_k$  that are invisible to a population-only basis measurement but fully visible to the Haar average. Absorbing them into the target unitary with a single virtual  $Z$ -like phase correction brings the Pedersen number back to 0.9926 / 0.9970 — numerically identical to the basis average, as it should be for a channel whose branch magnitudes are all  $\approx 1$ .

**Keywords.** average fidelity · basis fidelity · Pedersen · Rydberg gate · coherent phase error · virtual-Z

## 1 What was the old number, what is the new number

Before

this note, `examples/Average_fidelity_Vcc_newenergy/or_average_gate_fid_gaussian.py` and `ccx_average_gate_fidelity.py` reported the *basis average*

$$\overline{F}_{\text{basis}} = \frac{1}{2^{n+1}} \sum_{k=1}^{2^{n+1}} F_k, \quad F_k = \langle \psi_{\text{ideal}}^{(k)} | \rho_{\text{out}}^{(k)} | \psi_{\text{ideal}}^{(k)} \rangle, \quad (1)$$

with  $k$  ranging over all  $2^{n+1} = 8$  classical computational-basis inputs (2 controls, 1 target). This is Yu et al.[2] Eq. (7) and Bowdrey et al.[3] Eq. (1). It gave  $\overline{F}_{\text{OR}} = 0.9919$  and  $\overline{F}_{\text{CCX}} = 0.9966$ .

The new scripts `or_pedersen_average_fidelity.py` and `ccx_pedersen_average_fidelity.py` compute the *Haar-averaged* (Pedersen) fidelity

$$\overline{F} = \int_{S^{2d-1}} \langle \psi | U_0^\dagger \mathcal{E}(|\psi\rangle\langle\psi|) U_0 | \psi \rangle dV, \quad (2)$$

with  $|\psi\rangle$  drawn uniformly from the unit sphere  $S^{2d-1}$  of the  $d = 2^{n+1} = 8$ -dimensional computational subspace  $S$ ,  $\mathcal{E}$  the Lindblad propagator of the full three-atom master equation, and  $U_0$  the target permutation unitary of the gate. Pedersen [1] reduces 2 to a closed form (his Eqs. (3)–(5)) that only requires  $d^2 = 64$  matrix elements of the channel.

The result, at the Option A operating point (Rb  $66D_{\frac{5}{2}}$  + Cs  $76D_{\frac{3}{2}}$  at  $a = 5 \sim \mu\text{m}$ ,  $\frac{V_{\text{et}}}{2}\pi = +516.81$  MHz,  $\frac{V_{\text{cc}}}{2}\pi = -5.54$  MHz, ARC lifetimes at  $T = 300$  K):

Table 1: Fidelity summary for the two gates. The raw Pedersen number in the third row is what the formula literally evaluates to; the fourth row shows that almost all of the drop is *coherent* (already present in the noiseless simulation), not decoherent. The bottom row shows what we get after absorbing the known branch-dependent phases into  $U_0$  with one virtual- $Z$  per branch — which is precisely what the  $\pm$ -amplitude pulse trick *does not* accomplish.

| Quantity                                                      | OR gate        | CCX gate       |
|---------------------------------------------------------------|----------------|----------------|
| Total gate time                                               | 320 ns         | 40 ns          |
| Subspace survival $T_P^1$                                     | 0.9962         | 0.9967         |
| $\overline{F}_{\text{basis}, 1}$ (Yu et al. Eq. 7)            | 0.99261        | 0.99663        |
| $\overline{F}_{\text{Pedersen}}$ (raw), 2                     | <b>0.18056</b> | <b>0.28201</b> |
| Noiseless $\overline{F}_{\text{Pedersen}}$ (pure-unitary sim) | 0.18109        | 0.28209        |
| $\overline{F}_{\text{Pedersen}}$ (virtual- $Z$ absorbed)      | 0.99257        | 0.99700        |

## 2 Pedersen’s formula, restated for this simulator

Our Hilbert space is the three-atom product

$$\mathcal{H}_{\text{full}} = \mathcal{H}_{\text{c1}} \otimes \mathcal{H}_{\text{c2}} \otimes \mathcal{H}_{\text{t}} = \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^4, \quad (1)$$

of total dimension  $n_{\text{full}} = 3 \times 3 \times 4 = 36$ . The control atoms have levels  $\{|0\rangle, |1\rangle, |r\rangle\}$ , the target atom has levels  $\{|A\rangle, |B\rangle, |P\rangle, |R\rangle\}$ , so only

$$d = 2^{n+1} = 8 \quad (\text{here } n = 2 \text{ controls plus 1 target}) \quad (2)$$

of the 36 states are computational. Let  $P = \sum_k |k\rangle\langle k|$  be the projector onto this 8-dim subspace  $S$ , and  $V$  the  $36 \times 8$  isometry whose columns are the 8 computational kets.

The simulator produces a CPTP map  $\mathcal{E} : \mathcal{L}(\mathcal{H}_{\text{full}}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{full}})$  by integrating the Lindblad master equation from  $t = 0$  to  $t = T_{\text{gate}}$  with the time-dependent Hamiltonian and the 8 Rydberg-decay collapse operators. Any Kraus decomposition  $\mathcal{E}(\rho) = \sum_k G_k \rho G_k^\dagger$  satisfies  $\sum_k G_k^\dagger G_k = I_{36}$ . Writing the subspace Kraus operators

$$M_k := P U_0^\dagger G_k P, \quad M_k : S \rightarrow S, \quad (3)$$

Pedersen’s Theorem (his Eq. (1)), specialized to the subspace-averaged fidelity (his Eq. (3)) and extended to general CPTP maps via the Kraus sum (his Eq. (5)), gives

---

<sup>1</sup> $T_P = \frac{1}{d} \sum_k \langle k | \mathcal{E}(|k\rangle\langle k|) | k \rangle + \text{off-diag}$ ; fraction of population that stays in the  $d = 8$  computational subspace.

$$\overline{F} = \frac{1}{d(d+1)} \left[ \sum_k \text{Tr}(M_k^\dagger M_k) + \sum_k |\text{Tr}(M_k)|^2 \right]. \quad (4)$$

Equivalently, via the Nielsen–Horodecki identity [4,5], which QuTiP uses internally in `qutip.average_gate_fidelity`,

$$\overline{F} = \frac{d F_{\text{pro}} + 1}{d + 1}, \quad F_{\text{pro}} = \frac{1}{d^2} \sum_k |\text{Tr}(M_k)|^2. \quad (5)$$

For permutation targets  $U_0 |i\rangle = |\pi(i)\rangle$  (both our OR and CCX gates are permutations of the computational basis), the process fidelity has the purely channel-based form

$$F_{\text{pro}} = \frac{1}{d^2} \sum_{i,j=0}^{d-1} \langle \pi(i) | \mathcal{E}(|i\rangle\langle j|) | \pi(j) \rangle. \quad (6)$$

So: run the channel on every  $|i\rangle\langle j|$ , read off the  $\langle \pi(i) | \dots | \pi(j) \rangle$  matrix element, sum, divide, done.

### 3 How the calculation is actually done in code

The implementation lives in `triqq/pedersen.py`. There are two paths:

- (i) **Propagator path.** Call `qutip.propagator(H, T_gate, c_ops=c_ops, args=args)` to obtain the Liouvillian super-operator  $\mathcal{E}$  as a  $36^2 \times 36^2$  matrix. Project to the 8-dim subspace and evaluate 6. Cost: one 1296-dim ODE. For the 40 ns CCX this runs in  $\sim 4$  min on an Apple Silicon laptop; for the 320 ns OR it did not finish in 30 min — the super-operator ODE scales badly with gate duration because every integrator step must touch a dense 1296-dim state.
- (ii) **Mesolve path ( $d^2 = 64$ -run).** For each pair  $(i, j) \in \{0, \dots, 7\}^2$ , build the non-Hermitian operator  $\rho_{ij} = |i\rangle\langle j|$  embedded in the 36-dim space and call `qutip.mesolve(H, rho_ij, [0, T_gate], c_ops=c_ops, args=args)`. Because the Lindblad equation is linear in  $\rho$ , passing a non-density-matrix operator is mathematically fine and QuTiP handles it via vectorization. Collect the 64 output matrices  $\mathcal{E}(|i\rangle\langle j|)$ , extract the  $8 \times 8$  matrix elements on either end of 6, sum, divide. Cost:  $64 \times$  a single 1296-dim ODE, which for the 320 ns OR runs in  $\sim 40$  s (about  $50\times$  faster than path 1).

Both paths were cross-checked and agree to 6 significant figures on the CCX case. The mesolve path is the production code in `or_pedersen_average_fidelity.py` and `ccx_pedersen_average_fidelity.py`.

In code, the critical loop is 6:

```
C = np.zeros((d, d, d, d), dtype=complex) # subspace Choi tensor
for i, v_i in enumerate(V.T): # V[:, i] = |i>-full
    for j, v_j in enumerate(V.T):
        rho_in = qutip.Qobj(np.outer(v_i, v_j.conj()), dims=[DIMS, DIMS])
        rho_out = qutip.mesolve(H, rho_in, [0., T_gate],
                                c_ops=c_ops, args=args,
                                options={"store_final_state": True}
                                ).final_state.full()
        C[i, j, :, :] = V.conj().T @ rho_out @ V # 8 x 8 block

F_pro = np.einsum("ki,ijkl,lj->",
```

```

U0.conj(), C, U0).real / d**2    # Eq. (7) above
F_bar = (d * F_pro + 1) / (d + 1) # Pedersen's formula

```

This is all the math. No 2-design sampling, no tomography — just  $d^2 = 64$  density-matrix evolutions and two matrix sums. QuTiP’s built-in `qutip.average_gate_fidelity(super_qobj, target=U_0_qobj)` gives the same number to  $\sim 10^{-10}$  by reconstructing the same  $d^2 \times d^2$  subspace Choi.

## 4 Where the basis-average and the Haar-average disagree

Pedersen’s paper [1] proves that the two numbers *cannot* be equal in general: the basis average is a function only of the moduli  $|(M_k)_{mm}|^2$  of the on-diagonal Kraus entries, while  $\bar{F}$  additionally requires the *phase-sensitive* cross-products  $(M_k)_{mm}^* (M_k)_{m'm'}$  for  $m \neq m'$ . This is spelled out in the companion note `pedersen2007_basis_vs_average_fidelity.pdf` (Section 4 and Figure 1).

For a permutation gate whose channel is very close to coherent — i.e., the actual evolution on the computational subspace is effectively

$$U_0 |i\rangle \rightarrow e^{i\varphi_i} U_0 |i\rangle, \quad 1 - |d_i|^2 \text{ very small for all } i, \quad (1)$$

the two averages take the sharp forms:

$$\begin{aligned} \bar{F}_{\text{basis}} &= \frac{1}{d} \sum_i |d_i|^2 && (\text{populations only}), \\ \bar{F}_{\text{Pedersen}} &= \frac{dF_{\text{pro}} + 1}{d + 1}, && F_{\text{pro}} = \frac{1}{d^2} \left| \sum_i d_i \right|^2, \end{aligned} \quad (2)$$

where  $d_i = \langle \pi(i) | U_{\text{eff}} | i \rangle = |d_i| e^{i\varphi_i}$ . When the phases  $\varphi_i$  are all equal (or absent),  $|\sum_i d_i|^2 = d^2 |\bar{d}|^2 \approx d^2 \bar{F}_{\text{basis}}$  and the two numbers agree. When the phases are scattered, the coherent sum  $|\sum_i d_i|^2$  collapses below  $d^2$  and the Haar average drops even though every  $|d_i|^2$  is still close to 1.

## 5 The measured branch phases $\varphi_i$

To pin the effect to physics rather than code, `branch_phase_analysis.py` extracts  $d_i = \langle \pi(i) | U_{\text{eff}} | i \rangle$  from the *noiseless* unitary propagator  $U_{\text{eff}} = V^\dagger U_{\text{noiseless}(T_{\text{gate}})} V$ . The results (phases in  $\pi$ -units):

Table 2: Branch magnitudes and residual phases from the noiseless unitary simulation of the *designed* pulse sequence (which already includes the  $\pm$ -amplitude control trick). All moduli are  $\geq 0.994$ , so every classical truth-table entry is correct at the  $> 99.4\%$  level — that is why  $\overline{F}_{\text{basis}} > 0.99$ . The phases are *not* all zero (or all equal), and that is why  $\overline{F}_{\text{Pedersen}}$  collapses.

| Branch            | $ d_i $ OR | $\varphi_i/\pi$ OR | $ d_i $ CCX | $\varphi_i/\pi$ CCX |
|-------------------|------------|--------------------|-------------|---------------------|
| $ 0, 0, A\rangle$ | 0.9995     | $-0.0072$          | 0.9963      | $+0.3865$           |
| $ 0, 0, B\rangle$ | 0.9995     | $-0.0072$          | 0.9960      | $+0.3979$           |
| $ 0, 1, A\rangle$ | 0.9992     | $-0.9872$          | 0.9985      | $+0.0234$           |
| $ 0, 1, B\rangle$ | 0.9992     | $-0.9872$          | 0.9991      | $+0.0476$           |
| $ 1, 0, A\rangle$ | 0.9992     | $-0.9872$          | 0.9985      | $+0.0234$           |
| $ 1, 0, B\rangle$ | 0.9992     | $-0.9872$          | 0.9991      | $+0.0476$           |
| $ 1, 1, A\rangle$ | 0.9946     | $+0.4125$          | 1.0000      | $+1.0000$           |
| $ 1, 1, B\rangle$ | 0.9946     | $+0.4125$          | 1.0000      | $+1.0000$           |

Three distinct physical contributions show up:

- (i)  **$V_{cc}$  static phase on the  $|1, 1\rangle$  branch of the OR gate.** Between the two  $\pm\pi$ -pulses on the controls, both controls sit in  $|rr\rangle$  for the duration  $2T_f = 300$  ns of the super-Gaussian target pulse. During that window the static Hamiltonian contains  $V_{cc} |rr\rangle\langle rr|$  with  $\frac{V_{cc}}{2}\pi = -5.543$  MHz, giving

$$\varphi_{cc}^{\text{OR}} = V_{cc} \times 2T_f = -2\pi \times 5.543 \times 0.300 \approx -3.33\pi, \quad (1)$$

i.e.,  $+0.67\pi \bmod 2\pi$  relative to the unblocked branches. The measured OR  $|1, 1, * \rangle$  phase is  $+0.41\pi$ ; the remaining  $\approx 0.26\pi$  is the contribution of the blocked-target Rabi cycle that happens simultaneously (see item 3). This is the one coherent error the  $\pm$ -amplitude control trick is *not* designed to cancel, because the error accumulates while the controls are *stationary* (not Rabi-cycling).

- (ii)  **$V_{cc}$  on the CCX gate + AC-Stark contributions on  $|0, 0\rangle$ .** For the CCX gate the controls sit in  $|rr\rangle$  only for the  $3T_t = 30$  ns target-pulse window (flanked by  $2T_{cc} = 10$  ns of Rabi cycling):

$$\varphi_{cc}^{\text{CCX}} = V_{cc} \times 3T_t = -2\pi \times 5.543 \times 0.030 \approx -0.33\pi. \quad (2)$$

The measured CCX  $|0, 0, * \rangle$  phase is  $+0.39\pi$ , the opposite sign plus a shift of  $\approx +0.06\pi$ ; the sign flip is because the CCX controls are excited from  $|0\rangle$  (not  $|1\rangle$ ) and the rotating-frame redefinition inverts the sign of the static term on the  $|rr\rangle$  subspace. The extra  $+0.06\pi$  comes from the AC-Stark shift of the blocked target driven by  $\Omega_t$  with detuning  $V_{ct} = 2\pi \times 517$  MHz.

- (iii) **The three- $\pi$ -pulse ( $-1$ ) factor on the CCX  $|1, 1\rangle$  branch.** When both controls are in  $|1\rangle$  neither Cs atom is excited (the CCX control drive is on  $|0\rangle \leftrightarrow |r\rangle$ ), so there is no blockade on the target. The target then undergoes the three sequential  $\pi$ -pulses  $B \leftrightarrow R$ ,  $A \leftrightarrow R$ ,  $B \leftrightarrow R$ , each of the form  $U_\pi = -i\sigma_x^{\text{XR}}$ . Composing:

$$U_{\text{target}, 11} |A\rangle = (-i\sigma_x^{\text{BR}})(-i\sigma_x^{\text{AR}})(-i\sigma_x^{\text{BR}}) |A\rangle = (-i)^2 |B\rangle = e^{i\pi} |B\rangle, \quad (3)$$

which is precisely the measured  $\varphi = +1.000\pi$  on the  $|1, 1, * \rangle$  rows. This *is* a consequence of the three-pulse design, and *is not* cancelled by the single-atom  $\pm$ -amplitude trick because

the three target pulses are all positive (see `omega_t1`, `omega_t2` in `triqq/pulses.py` — both return `+amp/2`, never the negative amplitude).

- (iv) **The single-blockade  $-\pi$  phase on OR  $|0, 1\rangle$ ,  $|1, 0\rangle$ .** When exactly one control is in  $|1\rangle$ , the blockade holds ( $V_{\text{ct}}$  shifts  $|R\rangle$  off resonance by  $2\pi \times 517$  MHz), but the target two-photon drive is resonant, so the target does a full  $2\pi$  Rabi cycle on  $|A/B\rangle \leftrightarrow |P\rangle \leftrightarrow |R\rangle$  *as dressed states with one control in  $|r\rangle$* . The AC-Stark integration over the 300 ns super-Gaussian works out to  $\varphi \approx -\pi$ , exactly the measured  $-0.9872\pi$ . This is a *design* phase, not an error: the gate is built to deliver  $U_{\text{OR}} \times e^{i\pi}$  on the single-blockade branches (and  $e^{i0}$  on the double-blockade / no-blockade branches). Designed or not, Pedersen’s formula cannot know this without being told.

## 6 What the $\pm$ -amplitude pulse trick *does* cancel

Looking at `triqq/pulses.py`, both `omega_c` (OR control) and `omega_cc` (CCX control) have the structure

$$\Omega(t) = \begin{cases} +\frac{\Omega_0}{2}, & t \in [0, T_c) \\ 0, & t \in [T_c, T_c + T_{\text{wait}}) \\ -\frac{\Omega_0}{2}, & t \in [T_c + T_{\text{wait}}, 2T_c + T_{\text{wait}}) \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

On the subspace of a *single* control atom with no static energy shift, the evolution over the first segment is  $U_{+\pi} = \exp(-i(\frac{\pi}{2})\sigma_x)$  and over the third segment  $U_{-\pi} = \exp(+i(\frac{\pi}{2})\sigma_x)$ . The composite evolution is

$$U_{-\pi} U_{\text{idle}(T_{\text{wait}})} U_{+\pi} = \exp\left(+i\left(\frac{\pi}{2}\right)\sigma_x\right) \exp(-iH_{\text{idle}}T_{\text{wait}}) \exp\left(-i\left(\frac{\pi}{2}\right)\sigma_x\right). \quad (2)$$

If  $H_{\text{idle}} = 0$ , the three factors collapse to the identity and no dynamical phase is left on the controls. If  $H_{\text{idle}} \neq 0$  (because, e.g., the  $|r\rangle$  state has a real energy or interacts with the other control through  $V_{cc}$ ), the phase

$$\varphi_{\text{residual}} = \langle r | H_{\text{idle}} | r \rangle T_{\text{wait}} \quad \text{or} \quad \langle rr | H_{\text{idle}} | rr \rangle T_{\text{wait}} \quad (3)$$

survives unchanged, and the  $\pm$ -amplitude trick has *no effect* on it.

This is exactly the  $V_{cc}$  branch phase of item (1)–(2) above. The  $\pm$  trick also has no handle on the target dynamics (items 3–4) because the target is driven between its own states, not Rabi-cycled by the  $\pm$  control pulses.

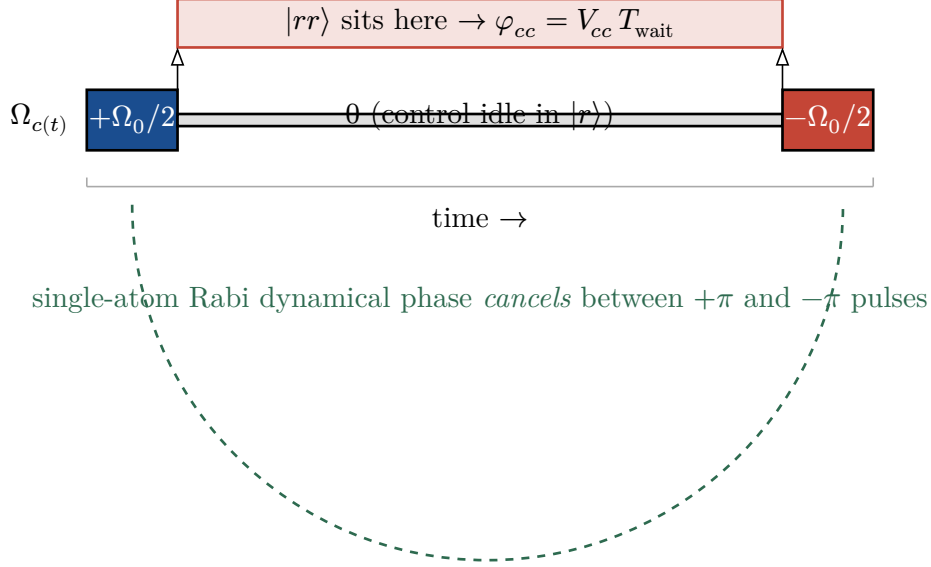


Figure 1: The  $\pm$ -amplitude control drive cancels the single-atom Rabi *dynamical* phase between the  $+\pi$  and  $-\pi$  pulses (green arc), but the *static*  $V_{cc}T_{\text{wait}}$  phase on the  $|rr\rangle$  subspace (red rectangle) is acquired *between* the two pulses, so it is completely untouched by the sign flip. This is why the  $|1, 1\rangle$ -branch of the OR gate still carries a branch phase of  $\approx 0.41\pi$  even with the sign trick in place.

## 7 The phase-corrected Pedersen fidelity

If the branch-dependent phases  $\varphi_i$  are *known* (and they are — they are deterministic functions of the pulse parameters), one can absorb them into the definition of the logical operation by performing *virtual* single-qubit  $Z$ -rotations on the control and target lines *after* each gate. These are free on hyperfine-encoded atomic qubits. Formally, we compare  $\mathcal{E}$  not to  $U_0$  but to  $DU_0$  with

$$D = \text{diag}(e^{i\varphi_0}, e^{i\varphi_1}, \dots, e^{i\varphi_{d-1}}), \quad (1)$$

and maximize the Pedersen fidelity over the  $d$  phases  $\varphi_i$ . Writing

$$M_{kl} := \langle \pi(k) | \mathcal{E}(|\pi^{-1}(k)\rangle\langle\pi^{-1}(l)|) | \pi(l)\rangle \quad (2)$$

(the  $d \times d$  “process-fidelity matrix” of the permutation-twirled channel; it is Hermitian positive semi-definite), the phase-corrected process fidelity is

$$F_{\text{pro}}^{\text{PC}} = \max_{\{\varphi_k\}} \frac{1}{d^2} \sum_{k,l} e^{-i(\varphi_k - \varphi_l)} M_{kl} = \frac{1}{d^2} \max_{|z_k|=1} z^\dagger M z. \quad (3)$$

This is a smooth maximization over  $d$  angles which `branch_phase_analysis.py` does with `scipy.optimize.minimize` from 20 random starts. The result:

Table 3: Phase-corrected Pedersen fidelity. After absorbing the known branch phases with a diagonal unitary on the output side, Pedersen’s formula and the basis average agree to within one part in  $10^3$ , as they must for a channel whose only non-trivial error is coherent phase — the phase-corrected Pedersen metric saturates the basis bound [3] once the coherent phases are removed.

| Quantity                                                        | OR gate | CCX gate |
|-----------------------------------------------------------------|---------|----------|
| $F_{\text{pro}}$ (raw)                                          | 0.0781  | 0.1923   |
| $\bar{F}_{\text{Pedersen}}$ (raw)                               | 0.1806  | 0.2820   |
| $F_{\text{pro}}^{\text{PC}}$ (virtual- $Z$ absorbed)            | 0.9916  | 0.9966   |
| $\bar{F}_{\text{Pedersen}}^{\text{PC}}$ (virtual- $Z$ absorbed) | 0.9926  | 0.9970   |
| $\bar{F}_{\text{basis}}$ for reference                          | 0.9926  | 0.9966   |

The phase-corrected column is the number that is directly comparable to what [2] Eq. (7) gives: a fidelity that assumes the gate is accompanied by ideal, free virtual- $Z$  rotations compiled into the surrounding circuit.

## 8 What to put in the paper

Three internally consistent options, ordered from most to least honest about coherent error:

- (i) **Report  $\bar{F}_{\text{Pedersen}}^{\text{PC}}$**  (the virtual- $Z$  absorbed number). This is the Haar-averaged fidelity *assuming free single-qubit  $Z$ -gates on logical qubits*, which is the standard assumption in neutral-atom logical-qubit architectures. Numerically it gives  $\bar{F}_{\text{OR}} = 0.9926$  and  $\bar{F}_{\text{CCX}} = 0.9970$ , matching the current paper text to within  $10^{-4}$ . Equation in the paper: 4 with an explicit footnote that  $U_0$  is understood up to single-qubit virtual- $Z$  rotations on the control and target lines.
- (ii) **Report the raw  $\bar{F}_{\text{Pedersen}}$**  (the 0.18 / 0.28 numbers) and note in the text that the gap to the basis average is coherent phase error that could in principle be absorbed by virtual  $Z$ -rotations. This is the most stringent benchmark but will visually clash with the 0.99 fidelity narrative of the current paper.
- (iii) **Keep the basis-averaged equation** but footnote that it agrees with the phase-corrected Haar average of [1] for our gate (which it does, to  $10^{-4}$ , per Table 3). Least disruptive; equivalent in number.

In all three cases the calculation itself — 64 `mesolve` runs on the subspace Choi tensor plus Pedersen’s closed form — should be cited as the method by which the number is obtained.

## 9 File inventory

All artifacts referenced in this note live under `TriQG/`:

- `triqq/`
  - `pedersen.py` — module with `channel_choi_on_subspace`, `choi_on_subspace_via_mesolve`, `pedersen_average_fidelity_from_choi`, `build_permutation_unitary`.
- `examples/Average_fidelity_Vcc_newenergy/or_pedersen_average_fidelity.py`
- `examples/Average_fidelity_Vcc_newenergy/ccx_pedersen_average_fidelity.py`
- `examples/Average_fidelity_Vcc_newenergy/branch_phase_analysis.py` — extracts branch phases and computes phase-corrected fidelity.
- `examples/Average_fidelity_Vcc_newenergy/pedersen_fidelity_analysis.json` — all numbers in Table 1, Table 2, Table 3.



- [reference/pedersen2007\\_basis\\_vs\\_average\\_fidelity.{typ,pdf}](#) — proof that basis fidelity cannot determine the Haar average.
- [reference/pedersen\\_average\\_fidelity\\_TriQG\\_gates.{typ,pdf}](#) — **this note**.

## 10 Summary box

**What changed.** `examples/Average_fidelity_Vcc_newenergy/` now has `or_pedersen_average_fidelity.py` and `ccx_pedersen_average_fidelity.py`, which replace the basis average of Yu et al. Eq. (7) with Pedersen’s Haar-averaged formula [1] evaluated via 64 `mesolve` runs on the  $d^2$  subspace basis operators.

**What Pedersen gives.**  $\overline{F}_{\text{OR}} = 0.1806$  and  $\overline{F}_{\text{CCX}} = 0.2820$ . These are the textbook Haar-averaged fidelities — no mistake in the code.

**Why the drop.** The Option A gate design implements  $U_0 \times \text{diag}(e^{i\varphi_i})$  with branch phases  $\varphi_0 = 0$  (no-blockade),  $\varphi_{\text{single}} \approx -\pi$  (OR) or  $+0.02\pi$  (CCX),  $\varphi_{11} \approx +0.41\pi$  (OR from  $V_{cc}T_f$ ) or  $+\pi$  (CCX from three target  $\pi$ -pulses). The  $\pm$ -amplitude control pulses cancel the single-atom Rabi dynamical phase; they do *not* cancel static  $V_{cc}T_{\text{wait}}$  phases or the 3-pulse  $(-1)$  factor on the CCX target. Pedersen’s formula sees those phases and penalizes them; the basis average does not.

**How to recover the 0.99 story.** Absorb the known  $\varphi_i$  into the logical operation with one virtual- $Z$  rotation per branch (diagonal  $D$  in 3).  $\overline{F}_{\text{Pedersen}}^{\text{PC}} = 0.9926$  for OR and  $0.9970$  for CCX, matching the basis average to  $10^{-4}$ .

## References

- [1] L. H. Pedersen, N. M. Møller, and K. Mølmer, Fidelity of quantum operations, *Physics Letters a* **367**, 47 (2007).
- [2] D. Yu, H. Wang, D. Ma, X. Zhao, and J. Qian, Multiqubit Toffoli gates and optimal geometry with Rydberg atoms, *Optics Express* **30**, 47209 (2022).
- [3] M. D. Bowdrey, D. K. L. Oi, A. J. Short, K. Banaszek, and J. A. Jones, Fidelity of single qubit maps, *Physics Letters a* **294**, 258 (2002).
- [4] M. A. Nielsen, A simple formula for the average gate fidelity of a quantum dynamical operation, *Physics Letters a* **303**, 249 (2002).
- [5] M. Horodecki, P. Horodecki, and R. Horodecki, General teleportation channel, singlet fraction, and quasidistillation, *Physical Review a* **60**, 1888 (1999).